

Airfoil Selection Trade Study

1. Methodology & Formulation

To determine the optimal airfoil among **E423**, **GOE165**, and **S1223** at a Reynolds number $Re = 200,000$, a Weighted Decision Matrix was implemented. Each aerodynamic parameter i is assigned a weight W_i based on mission priorities such that:

$$\sum_{i=1}^n W_i = 1.00 \quad (1)$$

Each airfoil j receives a discrete score $s_{ij} \in \{-1, 0, 1\}$ representing worst, average/baseline, and best performance, respectively. The total score S_j for airfoil j is calculated as:

$$S_j = \sum_{i=1}^n (W_i \cdot s_{ij}) \quad (2)$$

2. Extracted Aerodynamic Parameters

The baseline aerodynamic polar data calculated via XFLR5 ($Re = 200,000$) is summarized in Table 1.

Table 1: Extracted Aerodynamic Data ($Re = 200,000$)

Airfoil	$C_{L,\max}$	$C_{D,\min}$	$(L/D)_{\max}$	C_{m0} ($\alpha = 0^\circ$)	α_{stall}
E423	2.0338	0.01898	75.33	-0.2451	13.0°
GOE165	1.5447	0.01313	60.91	-0.1587	11.5°
S1223	2.2314	0.01778	71.81	-0.2716	13.0°

3. Weight Distribution & Evaluation Criteria

- **Maximum Lift Coefficient** ($C_{L,\max}$, $W_1 = 0.35$): Critical for payload capacity and take-off performance. Higher is better.
- **Maximum Lift-to-Drag Ratio** ($(L/D)_{\max}$, $W_2 = 0.25$): Defines aerodynamic efficiency and endurance. Higher is better.
- **Minimum Drag Coefficient** ($C_{D,\min}$, $W_3 = 0.20$): Minimizes parasitic drag during dash/cruise. Lower is better.
- **Zero-Lift Pitching Moment** (C_{m0} , $W_4 = 0.20$): Affects longitudinal stability and trim drag. Less negative is better.

4. Trade Study Decision Matrix

Table 2 presents the raw ratings, assigned weights, and weighted score calculations.

Table 2: Airfoil Trade Study Matrix

Parameter	Weight (W_i)	Scores (s_{ij})			Weighted Scores ($W_i \cdot s_{ij}$)		
		E423	GOE165	S1223	E423	GOE165	S1223
$C_{L,\max}$	0.35	0	-1	+1	0.00	-0.35	0.35
$C_{D,\min}$	0.20	-1	+1	0	-0.20	0.20	0.00
$(L/D)_{\max}$	0.25	+1	-1	0	0.25	-0.25	0.00
C_{m0}	0.20	0	+1	-1	0.00	0.20	-0.20
Total Score (S_j)	1.00	–	–	–	+0.05	-0.20	+0.15

5. Detailed Mathematical Calculations

The final weighted total scores are computed explicitly as follows:

$$S_{\text{E423}} = (0.35 \times 0) + (0.20 \times -1) + (0.25 \times 1) + (0.20 \times 0) = +\mathbf{0.05} \quad (3)$$

$$S_{\text{GOE165}} = (0.35 \times -1) + (0.20 \times 1) + (0.25 \times -1) + (0.20 \times 1) = -\mathbf{0.20} \quad (4)$$

$$S_{\text{S1223}} = (0.35 \times 1) + (0.20 \times 0) + (0.25 \times 0) + (0.20 \times -1) = +\mathbf{0.15} \quad (5)$$

6. Selection Conclusion

According to the trade study results, **S1223** obtains the highest total score ($S = +0.15$) and is recommended as the optimal choice due to its high lift capability ($C_{L,\text{max}} = 2.2314$).